

Letterboxd Movie Review Dataset

Data Collection and Sampling Methodology

Thesis: PENDING

Researcher: PENDING

Institution: PENDING

Report generated: 2026-09-27 08:52 UTC

Item	Value
Films in study	40
Target reviews per film	500
Target total	20,000
Language	English (Lingua)
Sampling	Simple Random Sampling, per film
Random seed	42
Review records ingested	0
Eligible English population	0
Reviews in final dataset	0
Films complete / insufficient / awaiting data	0 / 0 / 40

Table 0. Study summary (generated from the latest pipeline run).

1. Objective

The objective of this data-collection stage is to construct a dataset of English-language film reviews for 40 films, with 500 reviews per film (20,000 reviews in total), for subsequent text analysis in this thesis.

To limit sampling bias, reviews are not taken from the order in which the platform displays them (most recent, oldest, popular, most liked, or first page). Instead, for every film a sampling frame of all reviews accessible through the permitted data source is built first, the frame is cleaned and restricted to English, and a simple random sample is then drawn from the resulting eligible population with a fixed random seed.

At the time this report was generated (2026-09-27 08:52 UTC), no review data had yet been obtained through a permitted channel. The pipeline, templates and documentation are complete, and the dataset is produced by re-running the pipeline once permitted data is placed in data/raw/. All counts in this report therefore read zero; they are regenerated automatically on each run.

2. Data Source

NOTE: the data-source fields marked PENDING must be completed in config/data_source.json once the permitted source is confirmed (acquisition_route, source_description, permission_reference, collection_period, frame_definition, researcher, institution, thesis_title).

Platform: Letterboxd (<https://letterboxd.com>).

Acquisition route: PENDING - one of: (1) Official Letterboxd API with approved access; (2) written permission from Letterboxd for academic research; (3) a dataset whose licence/terms permit this use; (4) researcher-provided review data collected manually without bypassing any access control.

Source description: PENDING - describe exactly what data was obtained, from whom, and under what terms.

Permission / licence reference: PENDING - e.g. date and reference of Letterboxd's written permission / API approval e-mail, or dataset licence name and URL.

Collection period: PENDING - first and last date on which review data was obtained (YYYY-MM-DD to YYYY-MM-DD).

Sampling frame definition: PENDING - which reviews of each film the source made accessible (e.g. all public reviews returned by the approved API endpoint on the collection date).

Terms of Service review (2026-09-27). Letterboxd's Terms of Use (<https://letterboxd.com/legal/terms-of-use/>) state: "Except as explicitly authorized in these Terms, you must not employ any robot, spider, scraper, deep-link, or other automated data gathering or extraction tool, program, or algorithm to access, acquire, copy, or monitor any portion of the Service." Automated scraping of the Letterboxd website is therefore not permitted without explicit authorisation, and no scraping of letterboxd.com was performed for this study. Access to the Letterboxd API is by request only (api@letterboxd.com). Letterboxd states that it does not grant API access for data-analysis, visualisation or recommendation projects, for LLM/GPT-related use, or for private or personal projects. (<https://letterboxd.com/api-beta/>)

Consequently the study relies only on data obtained through a permitted route, in this order of preference: (1) the official API with approved access; (2) written permission from Letterboxd for

academic research; (3) a dataset whose terms allow this use; (4) review data supplied by the researcher; (5) manual, human-performed collection that does not bypass any protection. The collection software itself contains no web-access code: it only reads files that the researcher places in data/raw/, and each file must carry provenance (source and collection timestamp) or it is rejected.

Safeguards against bypassing access restrictions: no Cloudflare or CAPTCHA circumvention, no undetected/stealth browser automation, no browser-fingerprint spoofing, no proxy rotation to evade rate limits, no use of private or undocumented endpoints, and no circumvention of authentication or access control were used or are supported by the software.

3. Software and Tools

Python 3.11.15 was used for all processing. Package versions recorded at run time: pandas 3.0.6; numpy 2.4.6; openpyxl 3.1.5; lingua-language-detector 2.1.1; reportlab 5.0.1; pytest 9.1.1.

pandas was used for data handling, cleaning, de-duplication and sampling (`DataFrame.sample`); NumPy is the numerical back-end of pandas' random number generation; Lingua (`lingua-language-detector`) for language identification; openpyxl to write the Excel workbook; ReportLab to generate this PDF; pytest to run automated tests of the pipeline. Microsoft Excel (or any XLSX viewer) can be used to inspect the workbook. The code is plain Python scripts (`src/`) executed from the command line; no notebook is required.

4. Data Collection Procedure

The procedure follows the flow shown in Figure 1. Each step is implemented as a separate module in `src/` and executed in order by `src/pipeline.py`.

(a) Movie selection: the 40 films are listed in `config/movies.csv` with a stable identifier (M01-M40), title, year and Letterboxd slug. (b) Permitted data acquisition: raw review files are obtained outside the software through one of the permitted routes in Section 2. (c) Review population construction (`src/ingest.py`): all raw files are merged into one sampling frame, column names are standardised, each record is matched to a configured film, ratings and dates are normalised, a stable `review_id` is generated where the source provides none, and the username is replaced by a pseudonymous `reviewer_id`. (d) Cleaning (`src/clean_reviews.py`), (e) language detection and English filtering (`src/language_detection.py`), (f) de-duplication, (g) simple random sampling (`src/sampling.py`), and (h) export to CSV and XLSX (`src/export_csv.py`, `src/export_excel.py`), followed by automated validation (`src/validation.py`).

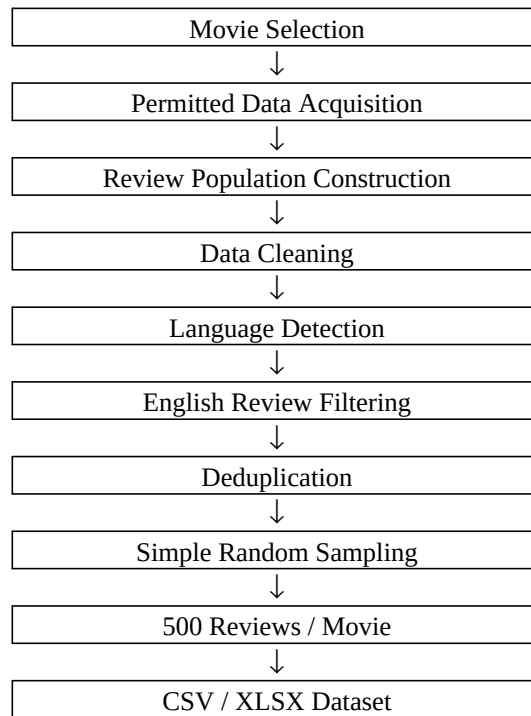


Figure 1. Data collection and sampling flow.

5. Random Sampling Method

Simple Random Sampling without replacement is used, applied separately to each film. The sample is drawn only after the eligible English review population of the film has been constructed; it is not a selection of the first 500 reviews shown by the website.

Before sampling, the eligible population is sorted by `review_id` so that the result does not depend on file order or on the platform's display order. The sample is then drawn with pandas:

`population.sample(n=500, random_state=42)`. Every review in the eligible population therefore has the same inclusion probability, $500/N$, where N is the size of that film's eligible population.

The fixed random seed (42) makes the sample reproducible: running the pipeline again on the same raw data produces exactly the same reviews, which the validation step verifies by re-drawing every sample. The seed is stored in the `sampling_seed` column of every record and in the Sampling Log.

If a film has fewer than 500 eligible English reviews, all eligible reviews are kept, no review is duplicated to reach 500, and the film is flagged `INSUFFICIENT_ENGLISH_REVIEWS` with its actual count in the Sampling Log.

```
eligible = eligible.sort_values('review_id')
sample = eligible.sample(n=500, random_state=42)
```

6. Language Identification

Language identification was performed before random sampling. Only reviews classified as English were retained in the eligible sampling population.

A single tool, Lingua (`lingua-language-detector`, version 2.1.1), was used for reproducibility. The detector was built from all languages supported by Lingua, in its default high-accuracy mode. For each review, Lingua's confidence values for every language were computed; the review was labelled with the

top-ranked language (ISO 639-1 code, stored in the language column) and its confidence value (language_confidence). Reviews without any identifiable language are labelled 'und'. A review is eligible when its top-ranked language is English; no additional confidence threshold was applied.

Lingua was chosen because it is designed for short and informal texts, which are common on Letterboxd. Very short reviews and reviews that mix languages remain harder to classify; see Section 10.

7. Data Cleaning

Invalid records were removed before language detection: empty texts and texts with fewer than 1 alphabetic character(s), such as emoji-only or punctuation-only reviews. Records that could not be matched to a configured film or lacked provenance were rejected at ingestion and listed in data/processed/rejected_rows.csv.

Duplicates were removed after English filtering, in this order: identical review_id; identical review URL; and exact identical text within the same film (compared after Unicode NFC normalisation and whitespace collapsing only). For each duplicate group the earliest-dated record is kept. Identical short texts under different films (e.g. 'masterpiece') are distinct reviews and are kept; their number is reported by validation.

Reviews were never removed because of their sentiment or rating. The original review text is preserved unchanged in the dataset. No lower-casing, stemming, lemmatisation, tokenisation, stop-word removal or spelling correction was applied; such text pre-processing for machine learning/NLP is a separate, later stage of the thesis.

8. Reproducibility

The following are recorded automatically on every run: Python version (3.11.15), package versions (Section 3, logs/run_metadata.json), random seed (42), collection dates (collected_at per record and collection_date per film), the Sampling Log (logs/sampling_log.csv) with population counts before and after each cleaning step, the list of every removed record with its reason (data/processed/removed_records.csv), and the validation report (logs/validation_report.txt). requirements.txt pins the package versions.

Given the same raw files in data/raw/, the command `python src/pipeline.py` regenerates the identical dataset, workbook, logs and this report.

9. Ethical Considerations

The data are used solely for non-commercial academic research. Only data obtained within the scope permitted by the data source are used; anti-bot mechanisms, Cloudflare, CAPTCHA, rate limits and authentication were not bypassed.

Personal data are minimised. Usernames are used only to derive a pseudonymous reviewer_id (SHA-256 hash, optionally keyed with a private salt through the LB_HASH_SALT environment variable) and are never written to any output file. No other profile information is collected. Because Letterboxd review URLs contain the username, the pipeline can blank the review_url column (--redact-urls) for any copy of the dataset that is shared outside the research team. Reviews are reported in aggregate; verbatim quotations in the thesis should be kept short and not attributed to identifiable

members.

If a salt is not used, an unsalted SHA-256 hash of a public username can be re-identified by hashing candidate usernames; it is therefore pseudonymous, not anonymous, and the dataset should be stored and shared accordingly.

10. Limitations

The sampling population represents the set of reviews accessible through the permitted data acquisition method and should not necessarily be interpreted as the complete population of all Letterboxd reviews for the film. The sample is a simple random sample of that accessible population only; it is not claimed to be random with respect to all reviews on Letterboxd.

Where the platform-reported number of reviews for a film is known, it is listed as `total_reviews_available` next to the number actually retrieved, so the coverage of each sampling frame can be judged. Where it is not known, coverage cannot be quantified.

Letterboxd reviewers are not representative of all film audiences, and reviews are self-selected; findings describe Letterboxd reviewers, not viewers in general.

Automatic language identification makes errors, especially on very short or mixed-language reviews, so a small number of English reviews may be excluded and a small number of non-English reviews may be included. Removing emoji-only and empty reviews slightly shifts the population toward reviews containing words.

Reviews can be edited or deleted after collection; the dataset reflects their state on the collection date. Films with fewer than 500 eligible English reviews are reported as insufficient rather than padded, so per-film sample sizes may differ.

Only data obtained through a permitted route can be used; if Letterboxd does not grant access, the scale or composition of the dataset may differ from the original plan, and any such change must be reported.

Appendix A. Sampling Log

Per-film counts from logs/sampling_log.csv. Avail. = platform-reported number of reviews (n/r = not reported by the source); Retr. = records retrieved into the sampling frame; Invalid = empty/emoji-only removed; English = English after language identification; Dup. = duplicates removed; Eligible = final eligible population; Sample = reviews in the final dataset.

ID	Film	Avail.	Retr.	Invalid	English	Dup.	Eligible	Sample	Status
M01	The Godfather	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M02	Pulp Fiction	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M03	The Shawshank Redemption	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M04	Fight Club	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M05	The Matrix	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M06	Spirited Away	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M07	The Dark Knight	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M08	Inception	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M09	The Social Network	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M10	Drive	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M11	Her	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M12	Interstellar	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M13	Whiplash	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M14	Mad Max: Fury Road	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M15	La La Land	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M16	Moonlight	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M17	Get Out	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M18	Lady Bird	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M19	Call Me by Your Name	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M20	Hereditary	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M21	Spider-Man: Into the Spider-Verse	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M22	Parasite	n/r	0	0	0	0	0	0	AWAITING RAW DATA

ID	Film	Avail.	Retr.	Invalid	English	Dup.	Eligible	Sample	Status
M23	Midsommar	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M24	Joker	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M25	Little Women	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M26	Knives Out	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M27	Portrait of a Lady on Fire	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M28	Everything Everywhere All at Once	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M29	Aftersun	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M30	The Batman	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M31	Pearl	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M32	Oppenheimer	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M33	Barbie	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M34	Past Lives	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M35	Poor Things	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M36	Killers of the Flower Moon	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M37	Dune: Part Two	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M38	Challengers	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M39	The Substance	n/r	0	0	0	0	0	0	AWAITING RAW DATA
M40	Anora	n/r	0	0	0	0	0	0	AWAITING RAW DATA

Appendix B. Validation Results

Type	Check	Result	Detail
COMPLETENESS	Movies configured = 40	PASS	40 configured
COMPLETENESS	Movies completed (500 reviews each)	INCOMPLETE	0/40 complete; 0 insufficient; 40 awaiting raw data
COMPLETENESS	Reviews collected vs target 20,000	INCOMPLETE	0/20,000
INTEGRITY	No film exceeds 500 reviews	PASS	ok
INTEGRITY	COMPLETE films have exactly 500 reviews	PASS	ok
INTEGRITY	Non-English records in final dataset = 0	PASS	0
INTEGRITY	Duplicate review IDs = 0	PASS	0
INTEGRITY	Duplicate review URLs = 0	PASS	0

Type	Check	Result	Detail
INTEGRITY	Exact duplicate review texts within a film = 0	PASS	0
INFO	Identical texts appearing under different films	INFO	0 (kept: different films are different reviews, e.g. short texts such as 'masterpiece')
INTEGRITY	Missing review texts = 0	PASS	0
INTEGRITY	sampling_seed recorded (= 42) on every record	PASS	0 records with other/missing seed
INTEGRITY	Every record states its source	PASS	0
INTEGRITY	Every record states collected_at	PASS	0
INTEGRITY	No username column in final dataset	PASS	ok
INTEGRITY	Every film has a Sampling Log entry	PASS	40 entries

Appendix C. Dataset Schema (letterboxd_reviews_all.csv / Reviews sheet)

Column	Description
movie_id	Film identifier from config/movies.csv (M01-M40)
movie_title	Film title
movie_year	Release year
review_id	Source identifier, or RID-<SHA-256> of the URL/record when none is supplied
review_text	Original review text, unmodified
rating	Star rating 0.5-5.0 if given, else empty
review_date	Date the review was published/logged (YYYY-MM-DD)
language	ISO 639-1 code assigned by Lingua (always 'en' in the final dataset)
language_confidence	Lingua confidence value for that language (0-1)
review_url	Review URL (blank when --redact-urls is used)
reviewer_id	SHA-256 pseudonym of the username (optionally salted)
sampling_seed	Random seed used for sampling (42)
source	Permitted data source of the record
collected_at	UTC timestamp at which the record was obtained